

See It Again Before You Forget It.

Spaced Repetition · Ebbinghaus · Cepeda, Pashler & colleagues

"I knew all 28 pulse qualities on Sunday.
By Wednesday I remembered maybe twelve."

— That's not a memory problem. That's the forgetting curve doing exactly what it does.

In 1885, Hermann Ebbinghaus spent months memorizing lists of nonsense syllables and testing himself at intervals. What he found became one of psychology's most replicated findings: **memory decays exponentially after learning, and the rate of decay follows a predictable curve.** By one hour after studying, you've forgotten roughly 50% of new material. By one week, 75–80%. But every time you successfully retrieve the information just before forgetting it, you reset the curve at a higher baseline — and the next decay is slower.

The Research: Optimal Spacing Intervals

Cepeda, Pashler, Vul, Wixted, and Rohrer (2006) conducted the largest meta-analysis of spacing research ever assembled — 254 studies, 14,000 participants. Their findings: distributed practice (spaced across sessions) consistently outperforms massed practice (cramming) for long-term retention, with the optimal gap between study sessions scaling with how far in the future you need to remember the material.

254

studies synthesized in Cepeda et al. (2006) — one of the largest meta-analyses in cognitive psychology

10–30%

retention gain from spaced vs. massed practice on delayed tests (Cepeda et al., 2006)

1→3→7

days — the basic spacing sequence for a one-week exam. See it once, review in 1 day, review in 3 more days.

For a quiz one week away, optimal spacing looks like: **study today → review tomorrow → review in 3 days → quiz.** Three encounters, each one preventing a full decay, each one rebuilding the memory trace stronger than before. Cramming the night before produces one encounter with full decay in between sessions.

The Forgetting Curve — and How to Bend It

Each successful retrieval just before forgetting does three things: resets the decay back to 100% retention, **slows the rate of the next decay**, and strengthens the long-term storage trace. After four or five well-timed retrievals, recall becomes nearly automatic — the material is in long-term memory, not working memory. This is the difference between knowing something for a quiz and knowing it when a patient is in front of you at 11pm.

The spacing sweet spot: Review material at the point where you’ve forgotten maybe 20–30% of it — not when it’s fresh (too easy, no encoding benefit) and not when it’s completely gone (too hard, mostly re-learning from scratch). The slight effort of partial recall is the signal you’re in the right window.

Your Built-In Spaced Repetition System

You already have a spaced repetition system built into this course — you’re just not using it that way yet.

Course Element	Spaced Repetition Use
Pulse/Tongue Journal (50 entries)	Each time you take and record a pulse, you’re retrieving all 28 qualities against a real finding. Done across 9 weeks, this IS a spaced repetition schedule — if you spread entries out rather than batching them at the end.
Weekly PLAS (Prep Guide)	Section B (Activate) forces you to retrieve prior week material in service of the new week. This creates a natural spacing interval: content from last week reappears one week later.
Section F (Post-Lecture Synthesis)	Completing Section F within 24 hours is a retrieval event timed exactly at the first major decay point — roughly where the forgetting curve drops to 50%. This is the most important timing intervention you can make.
Practice Quiz	Take it immediately after class (Day 1), again 3 days later without looking at notes, again the day before the quiz. Three spaced retrievals that bracket the optimal decay window.
Cram Sheet	The cram sheet is not for cramming — it’s for the final spacing event the morning of the quiz, when all prior retrievals have already done the encoding work.

The journal is your long-game spaced repetition: Every pulse/tongue entry that goes in entries 1–25 should be spread across weeks 1–4, not batched before the midterm. A pulse entry in week 1, another in week 2 using the same quality, another in week 3 — that’s three retrievals with increasing spacing. Batching 25 entries in week 4 is cramming in journal form.

Practical schedule for pulse quality mastery:

Day 1 (after lecture): Section F brain dump + practice quiz cold

Day 2: Review wrong answers only + one journal entry

Day 4: Practice quiz again without notes

Day 6 (eve of quiz): Cram sheet only — the encoding is already done

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